



Black Hat USA 2026: Practical Linux Attack Paths and DFIR/Hunting v2.0

Training dates: August 1-4, 2026

Location: Las Vegas

Practical Linux Attack Paths and DFIR/Hunting v2.0 training has been created with a focus on realistic hands-on experience in analyzing the behavior of user space, kernel space, and eBPF techniques, covering recent privilege escalation, persistence, defense evasion, and C2/LM internals vs live, behavioral detection, hunting/DFIR tooling, and generic prevention strategies in the Linux ecosystem. Dive into the world of modular Linux attack paths, local and remote exploitation, process injection, process hiding, network tunneling/pivoting, data exfiltration, and syscall hooking techniques. Get hands-on experience on how Linux malware and US/KS/eBPF rootkits work in the well-prepared PurpleLabs Cyber Range. Analyze and modify source code, identify interesting behavior patterns in binaries and logs, focus on artifacts and events, determine which telemetry is needed to detect modern Linux threat actors, find ways to proactively validate and improve your EDR/SOAR/SIEM setup through step-by-step Linux adversary emulations, and eventually activate custom blocking policies using eBPF LSM C and Tetragon. On top of that, run your VMs' RAM acquisition "on click" and analyze memory images with Volatility Framework 3 at any stage of the training. Full scope, 100% modern Linux/K8S Purple Teaming structure, and only hands-on delivery style.

The hands-on session will be based on the PurpleLabs Cyber Range Environment, backed by the EDRmetry Matrix and the corresponding material of the Linux Attack, Detection, and Forensics v2.0 - Hands-on Purple Teaming Playbook course.

PurpleLABS is a dedicated, virtual infrastructure for detecting and analyzing an attacker's behavior in terms of techniques, tactics, procedures, and offensive tools used. The environment is to serve the purpose of constant improvement of competencies in the field of threat hunting and learning about current trends in red teaming vs direct detection (blue teaming).

PurpleLABS provides analytical interfaces to all relevant sources of data from individual systems and services available in virtual infrastructure (EDR/Runtime Security, Zeek, Suricata, OSQuery, syslog, WAF, Proxy, audit, eBPF, and more).

This course takes on an "Attack-Observe-Detect-Enforce" approach in a condensed, modular format:

- Emulate offensive technique
 - Check for alerts, find corresponding events in raw telemetry, and correlate with other data sources
 - Focus on forensic artifacts and behavioral profile/visibility of technique
 - Activate mitigation policies composed of baseline profile enforcers, anti-pivoting/anti-lateral movement, and live patching/anti-exploit probes
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Training overview:

- This training is about Linux offensive techniques in the format of ready-to-copy-and-paste blocks of code that you can use during your pen-testing/emulation engagements or detection coverage testing/EDR evaluations, with the possibility of chaining and creating advanced Linux custom attack paths.
- In today's Linux threat landscape, enterprises face increasingly sophisticated, targeted attacks. To effectively combat these threats, we must enhance our ability to detect malicious activity, inform threat-hunting processes, and understand attacker behavior.
- This training aims to show and describe in detail chunks of different Linux offensive executions we can usually find during real attack campaigns. There are a few main ideas and goals behind the course:
 - Teach you what offensive techniques exist in Linux, with mapping to the Linux Matrix Attack Framework
 - Provide ready-to-copy-and-paste blocks of code that you can use during your own penetration testing/emulation engagements, detection coverage testing, or EDR evaluations, with the possibility of chaining and creating advanced Linux custom attack paths
 - Provide step-by-step instructions about how to generate payloads, how to set up and operationally use C2 frameworks like Mythic, Sliver, or Merlin
 - Teach you how to handle true hands-on evaluation testing of modern Linux EDR
 - Explain what you should expect from modern Linux EDR products with a focus on the internals, capabilities, detection, and operational efficiency

- Introduce you to commercial Linux EDR solutions and open-source Runtime Security implementations as well.
 - Custom Prevention/Response Add-On:
 - Detection and Forensics tell you if the attacker is inside. Prevention and response are what you do ahead of the attack. Mass exploitation campaigns are running faster and at a greater scale than at any point in the history of enterprise Linux. In raw absolute numbers, the vulnerability exploitation problem is worse than it has ever been.
 - The Prevention/Response Add-on for every offensive lab scenario extends the base course with the other side of the kill chain -> stopping/blocking attacks before they complete.
 - Full scope, not vendor-dependent. Not signature-dependent. Grounded in the same low-level Linux internals that the base course teaches.
 - AI made every attack faster. The Prevention/Response Add-on section makes stopping them practical via custom kernel-level enforcement policies within at least 4 different stages:
 - Baseline Profiling
 - Profile Enforcing
 - Anti-Lateral Movement & Anti-Pivoting
 - Live Patching
 - During the training, we'll examine and learn how to execute custom EDR validation checks mapped to the MITRE ATT&CK™ Framework as evaluation checks. We are not going to focus on one specific vendor or tool. Thanks to the program of this course, you will be able to run your own evaluation tests against whatever EDR you have/want to have in production. The course content will also greatly expand your knowledge in the area of Linux threat hunting.
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Agenda:

- **Introduction:**
 - Welcome to the v2.0 party!
 - Goals / What to expect
 - Active Defense, PT, DE & Assume Breach
 - Linux DFIR Introduction
 - PurpleLabs Dashboard
 - General Course Flow Design
 - Private Virtual Machines
 - Shared Virtual Machines
 - PurpleLabs VM Robot Tool

- PurpleLabs Network Diagram
 - EDRmetry Matrix
 - AI Integration
 - Why Linux as a target?
 - Linux Threat Landscape
- **The architecture of the modern Linux Runtime/EDR engine:**
 - Core functionalities and key features
 - Visibility events/indexes/data sources
 - Syscalls, Kernel Functions, and Tracing
 - Detection/Prevention logic and rulesets → **based on Falco/Kunai/Tetragon**
 - Analytics / Query language
 - Triage and Forensics collections
 - Deployment and Integrations
 - **Defensive/DFIR Tooling:**
 - Host/Exploration of CLI tools
 - Host/Exploration of /proc
 - Host/Exploration of /sys
 - Host/Falco Runtime Security
 - Host/Kunai Runtime Security
 - Host/Tetragon Runtime Security
 - Host/Tetragon Custom Tracing Policies
 - Host/Custom eBPF C Prevention Techniques
 - Host/Elastic Security Agent
 - Host/Velociraptor Agent
 - Host/FleetDM OSquery Agent
 - Host/Sandfly Security
 - Host/go-journalctl
 - Host/auditd
 - Host/Linux IR Scripts - SecureProbes
 - Host/UAC
 - Host/Ghostscan
 - Host/Decloaker
 - Host/bpftrace
 - Host/LKRG
 - Host/dissect
 - Memory/RAM acquisition
 - Memory/Volatility3 Framework
 - Memory/Volatility2 Framework
 - Memory/mquire

- Research and emulation of Linux threats → The searchable playbook in attack-detection-hunting-prevention mode:
 - **Initial Access:**
 - EDR-T6261 - Remote UAF Exploitation - user
 - EDR-T6119 - Remote UAF Exploitation - root
 - EDR-T6354 - Remote UAF+Heap Overflow
 - EDR-T6062 - Kafka CVE-2023-25194
 - EDR-T6355 - Langflow API CVE-2025-3248
 - EDR-T6114 - ActiveMQ CVE-2023-46604
 - EDR-T6105 - Apache HTTP CVE-2021-41773
 - O-EDR-T6116 - Tomcat Upload Manager
 - EDR-T6110 - Solr Log4J JNDI CVE-2021-44228
 - EDR-T6233 - XZ Backdoor CVE-2024-3094
 - EDR-T6113 - Spring CVE-2022-22963
 - EDR-T6416 - React2shell - CVE-2025-55182
 - EDR-T6450 - K8S Ingress-NGINX CVE-2025-1974
 - EDR-T6464 - Tomcat Tribes EncryptInterceptor RCE - CVE-2026-34486
 - EDR-T6467 - Flask SSTI
 - **Discovery:**
 - EDR-T6225 - Execute "What Server"
 - EDR-T6040 - Execute LinPEAS from /dev/tcp
 - EDR-T6065 - /proc/PID/ Enumeration
 - EDR-T6251 - Process Snooping with pspy
 - EDR-T6280 - Find loaded eBPF programs/maps
 - EDR-T6218 - Linux VM Check via Hardware
 - EDR-T6204 - Read local file using curl
 - EDR-T6010 - Check my public IP
 - EDR-T6412 - DNS Reconnaissance
 - EDR-T6399 - Access to the AWS Metadata Service
 - **Defense Evasion:**
 - EDR-T6277 - Built-in System Tools Execution
 - EDR-T6041 - Execute binary listening from a hidden directory
 - EDR-T6039 - File Transfer to a hidden directory
 - EDR-T6138 - Bash HTTP GET data with /dev/tcp
 - EDR-T6173 - Hackshell + OpenSSL download
 - EDR-T6340 - Python HTTP POST and Exec
 - O-EDR-T6363 - Base64 Payload inside ZIP
 - EDR-T6370 - Sneaky_remap + Ptrace() Process Injection in Rust + SSL/TLS callback
 - EDR-T6439 - Malasada .so to Shellcode Loader

- EDR-T6127 - dd+/proc/PID/mem Injection
- EDR-T6256 - STOP/CONT Process Injection
- EDR-T6108 - ASM Injection over /proc/PID/mem
- EDR-T6037 - Python3 Fileless memfd_create
- EDR-T6188 - Fileless Execution with memexec
- EDR-T6171 - Easy Proc Name Masquerading
- EDR-T6038 - Proc Name Masq with exec
- EDR-T6140 - Proc Name Masq with prctl()
- EDR-T6345 - Proc Masq with mount NS
- EDR-T6046 - LD_PRELOAD Proc Hiding
- EDR-T6053 - mount --bind process hiding
- EDR-T6253 - eBPF Process Hider
- EDR-T6485 - Falco BPF Map Poisoning
- EDR-T6107 - LKM Fileless Remote Loading
- EDR-T6396 - LKM Hooking init_module
- EDR-T6293 - LKM Disabling SELinux
- EDR-T6282 - LKM Reset Yama ptrace_scope
- EDR-T6167 - BOF Loading with BOF-Stager
- EDR-T6067 - SSH notty session
- EDR-T6133 - File immutable with chattr
- EDR-T6089 - Bashrc File Hiding with ls Alias
- O-EDR-T6078 - Execute tools via PRoot BYOF
- EDR-T6455 - Fake TLS ClientHello DPI bypass
- EDR-T6025 - Encrypted ELF implant

- **Persistence:**

- EDR-T6395 - LKM LibZeroEvil r00tme
- EDR-T6394 - LKM Singularity Rootkit
- EDR-T6100 - LKM Char Device + LPE
- EDR-T6163 - LKM Reveng Rootkit
- EDR-T6023 - LKM Diamorphine Rootkit
- EDR-T6289 - LKM Ftrace Rootkit - Rebellion
- EDR-T6161 - LKM BDS Ftrace Hooking Rootkit
- EDR-T6154 - LKM Suterusu Rootkit
- EDR-T6155 - LKM KoviD Rootkit
- EDR-T6123.016 - Execute LKM
call_usermodehelper() on ICMP
- EDR-T6152 - eBPF Boopkit Rootkit
- EDR-T6325 - BPFDoor Backdoor
- EDR-T6454 - eBPF XDP UDP Backdoor
- EDR-T6151 - eBPF TripleCross Rootkit
- EDR-T6104 - SSHD Dummy Cipher Suite BYOT
- EDR-T6235 - LD_PRELOAD Re-adding SSH key
- O-EDR-T6327 - LD_PRELOAD Father Rootkit

- EDR-T6422 - bdvl - Patch Dynamic Linker
- EDR-T6423 - SSHD id_ed25519 Key Backdoor
- EDR-T6170 - Cap_setuid over LD linker
- EDR-T6139 - Python .pth Extensions
- O-EDR-T6407 - Shadow SUID binfmt_misc
- EDR-T6093 - Crontab root Backdoor
- EDR-T6431 - Supervisor persistence
- EDR-T6015 - Systemd Backdoor service
- EDR-T6179 - Udev+atd C2 persistence
- O-EDR-T6213 - Malicious RPM package
- EDR-T6164 - PAM Sneaky Backdoor
- EDR-T6144 - DNF Package Manager
- EDR-T6051 - Modify core_pattern
- EDR-T6347 - Nginx Shell Module
- O-EDR-T6017 - HTTPD mod_backdoor
- EDR-T6011 - PHP Webshells

- **Privilege Escalation:**

- EDR-T6109 - LPE Socket Command Injection
- EDR-T6360 - UAF Dirty Page Table LPE
- EDR-T6359 - UAF Cross-cache Dirty Pipe LPE
- EDR-T6184 - PATH Hijacking
- O-EDR-T6315 - Add SSH key via iptables-save
- EDR-T6335 - sudo chroot CVE-2025-32463
- EDR-T6147 - Docker Escape with socket+Chisel
- EDR-T6417 - Docker Escape with core_pattern
- EDR-T6187 - NFS SUID Escalation
- EDR-T6230 - pkexec CVE-2021-4034
- EDR-T6231 - DirtyPipe CVE-2022-0847 LPE
- O-EDR-T6177 - MySQL UDF Command Exec
- EDR-T6459 - Copy Fail - CVE-2026-31431 LPE
- EDR-T6462 - Fragnesia LPE
- EDR-T6463 - pidfd_getfd + ptrace - Read root files
- O-EDR-T6468 - DirtyClone CVE-2026-43503
- EDR-T6480+EDR-T6481+EDR-T6290 - Overwrite modprobe_path - physmap/virtual/sysfs
- EDR-T6457+EDR-T6051 - Modify core_pattern via sysfs/direct syscalls

- **Exfiltration:**

- EDR-T6120 - Python FTP Upload
- EDR-T6180 - SMB Data Exfiltration
- EDR-T6112 - NTP Data Exfiltration
- EDR-T6137 - HTTP PUT method + transfer.sh

- EDR-T6181 - Upload/download data over SSHFS
- O-EDR-T6135 - Upload data over WebDAV
- EDR-T6234 - pam_exec SSHD Exfiltration
- EDR-T6445 - PAM XOR VeilCreds
- O-EDR-T6103 - PAM creds over HTTP Post
- O-EDR-T6168 - ICMP_exfil + nping
- EDR-T6418 - LDAP Data Hiding - FreelPA
- EDR-T6342 - Exfil data with AWS S3

- **Command and Control:**

- EDR-T6123.023 - curlrevshell
- EDR-T6123.008 - Revshell openssl+/dev/fd/3
- EDR-T6123.004 - Revshell mkfifo+nc
- EDR-T6123.022 - Oneshell - echo and chmod
- EDR-T6123.015 - Revshell Python TLS
- O-EDR-T6126.002 - Sliver C2 MTLS
- EDR-T6126.013 - Sliver C2 HTTPS
- O-EDR-T6126.007 - Sliver C2 DNS
- O-EDR-T6126.011 - Sliver C2 TCP Pivots
- EDR-T6212 - Emp3r0r HTTP2 AES Stager C2
- EDR-T6317 - SOA/ECS DNS C2 Channel
- EDR-T6448 - ICMP Assembly C2
- EDR-T6115 - DNS Tunneling with dnscat2
- O-EDR-T6409 - Venom C2
- O-EDR-T6126.006 - Platypus C2
- EDR-T6126.004 - Merlin HTTP2 C2
- EDR-T6126.010 - Mythic C2 Deployment
- O-EDR-T6126.003 - Mythic C2 Thanatos ELF
- O-EDR-T6126.009 - Mythic C2 Medusa Python
- O-EDR-T6126.005 - Mythic C2 Poseidon ELF
- O-EDR-T6442 - Mythic C2 Dolios AWS S3
- O-EDR-T6148 - XOR shell_reverse_tcp Loader
- EDR-T6117 - UPX Reverse SSH server
- EDR-T6123.017 - Shell over HTTP streams
- EDR-T6328 - io_uring Agent

- **Credentials Access:**

- EDR-T6242 - eBPF Spy on PAM with python3
- EDR-T6199 - eBPF pamspy
- EDR-T6486 - eBPF httpdsby with python3
- EDR-T6012 - Sniff sshd with strace
- EDR-T6415 - ptrace() ssh-inject
- EDR-T6319 - Dump heap memory from Java

- EDR-T6466 - Cred_hunter - Cloud & CI/CD Credential Hunter

- **Impact:**

- O-EDR-T6063 - Ransomware C - lokpack
- O-EDR-T6252 - Crypto Mining CPU stress
- O-EDR-T6018 - Ransomware bash+openssl

- **Lateral Movement:**

- EDR-T6189 - Reverse SOCKS5 proxy
- EDR-T6357 - Chisel Reverse Socks Proxy
- EDR-T6255 - KCP - FRP Fast Reverse Proxy
- EDR-T6033 - SSH Linux Tunneling
- EDR-T6404 - mTLS Reverse SOCKS5
- O-EDR-T6387 - SOCKS5 over Tailscale
- EDR-T6392 - Cloudflared Tunneling
- EDR-T6131 - Hijacking SSH Client Session
- O-EDR-T6364 - Hijacking ssh-agent session
- EDR-T6057 - Execute Port Scanning

- **Kubernetes Add-on:**

- K8S Components
- K8S Deployment
- K8S gVisor Deployment
- Falco Deployment in K8S
- Tracee Deployment in K8S
- Kubernetes Threat Matrix
- Kubectl Fu
- K8S Simple Pod Deployment
- K8S Unauthenticated API Access
- K8S Static Hidden Pod
- K8S Pod Breakout
- K8S Mounted secrets
- K8S Etcd Dump
- K8S CronJob SSH Persistence
- K8S Pod Name Similarity
- K8S Grafana + PostgreSQL RCE
- K8S Vuln Web App
- K8S - EDR-T6105 - Apache HTTPD CVE-2021-41773
- K8S - EDR-T6464 - Tomcat Tribes CVE-2026-34486
- K8S - EDR-T6062 - Kafka CVE-2023-25194
- K8S - EDR-T6114 - ActiveMQ CVE-2023-46604
- K8S - EDR-T6467 - Flask SSTI
- K8S - EDR-T6355 - Langflow API CVE-2025-3248

- K8S - EDR-T6416 - React2shell - CVE-2025-55182
 - K8S - EDR-T6113 - Spring CVE-2022-22963
 - K8S - EDR-T6354 - Remote UAF+Heap Overflow
 - K8S Security Scanners
- DFIR/Hunting use cases in the modern Linux/Kubernetes environment
 - Real-life Linux APT behaviors and EDR bypass cases
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Values For Red/Blue/SecOps Teams:

- Understand the advantages and values of the purple teaming approach in the Linux red/blue ecosystem
 - Learn about the full scope of Linux offensive techniques, tools, and the newest community research 2024/2025/2026
 - Learn about different detection/response tools and techniques vs attacks
 - Learn how to hide effectively in the Linux OS and how to exfiltrate data in stealthy ways
 - Learn how to deploy and use C2, low-level rootkits, and see this reflected in the detection/DFIR tooling
 - Get code and command snippets ready to use during your red team and adversary operations/emulations
 - Get experience with Sigma Rules/Protections Artifacts for staying stealthier and improving your defense evasion skills at scale
 - See the effectiveness of the Detection tooling vs attack emulations
 - Learn about the full scope of Linux Detection/Forensics techniques, tools, and the newest community research
 - This knowledge will change the way you look at hardening and monitoring your Linux ecosystems
 - Recognize security-related enhancements in the modern Linux kernel
 - Understand current kernel components and programming interfaces used to compromise a system
 - Discover recommended open-source security solutions against actual hands-on attacks
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This knowledge will change the way you look at hardening and monitoring your Linux ecosystems:

- Learn about the full scope of Linux offensive techniques, tools, and the newest community research 2024/2025/2026
- Understand the advantages and values of the purple teaming approach in the Linux red/blue ecosystem
- Learn about different EDR/detection/response tools and techniques vs attacks
- Learn how to hide effectively in the Linux OS and how to exfiltrate data in stealthy ways
- Learn how to deploy and use C2, low-level rootkits, and see this reflected in the detection/DFIR tooling
- Get code and command snippets ready to use during your red team and adversary operations/emulations
- Understand the structures of advanced Linux attack paths, how they work, and how to protect against them
- Recognize security-related enhancements in the modern Linux kernel
- Understand current kernel components and programming interfaces used to compromise a system

General benefits:

An advanced hands-on PurpleLABS training laboratory has been created to present:

- The value of the Assume Breach approach and simulation of threats after getting early access to the target. (Discovery, C2, Lateral Movement, Persistence, Evasion, Exfiltration, Execution, Credential Access)
- The importance of Blue and Red team cooperation and how to effectively run hunting activities and write security notes.
- “Feel the network and systems” approach to getting and understanding the baseline behavior of devices, OS, and network.
- Different ways for playing with many important data sources, including Syslog, Kunai, Tetragon, Falco, Tracee, Yara, eBPF, Zeek, Suricata, OSQuery, memory dumps, and Full Packet Captures.
- How to run adversary simulations effectively, including the development of Attack Paths and Chain Attack scenarios by combining the attacker’s techniques, tactics, and procedures.

- Visibility, detection methods, and capabilities of well-recognized Hunting and Detection tools, including Elastic Security, Splunk, Moloch, OSQuery, Wazuh and Sandfly Security
 - Develop the team's analytical skills required to work in the SOC environment.
 - Increase awareness of the complexity and dependencies between the elements of the APT campaign and the areas of detection.
 - Understand what threat hunting is and why it is important.
 - Acquire skills related to generating suspicious events on the layer of the network and Linux operating systems, and methods of their detection.
 - Run a validation of the current security status of the organization's network and the risks involved.
 - Obtain knowledge on supplying/creating a complete SOC environment using open-source software.
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Student Requirements:

- Fundamentals of how Linux/K8S Architecture works are required
 - An intermediate level of Linux command-line syntax experience
 - Basic knowledge of TCP/IP network protocols
 - Offensive Security/Penetration testing experience will definitely be beneficial, but not required
 - Basic programming skills are a plus and are essential
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Target audience:

- Red and Blue team members
 - Security / Data Analytics
 - CSIRT / Incident Response Specialists
 - IT Security Professionals, Experts & Consultants
 - Network Security Engineers and Linux System Administrators
 - SOC members and SIEM Engineers
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Black Hat USA Registration link:

- <https://blackhat.com/us-26/training/schedule/index.html#practical-linux-attack-paths-and-dfirhunting-v20-50776>